



ELECT MAINT- OFF SITE Department
SIMHADRI

Note No :1



Ref:SMTTP/ELECT MAINT-/2023-24/BUSDUCT/432803

Date:

28/08/2023(12:06:30)

Subject:Schedule review of LMI/ OGN/OPS/ELEC/039-Guidelines on Busduct Maintenance and Overhauling

Subject: Schedule review of LMI/ OGN/OPS/ELEC/039-Guidelines on Busduct Maintenance and Overhauling.

LMI/ OGN/OPS/ELEC/039 on **Guidelines on Busduct Maintenance and Overhauling** is reviewed as per schedule and no changes are envisaged.

Review proposal of LMI is submitted for the kind approval.

Approving authority for above proposal is HEAD OF THE PROJECT.

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Found in line with OGN.

All related sections to review further

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Maintenance task list attached as Annexure-III

May please process the file further.

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NTPC Limited

SIMHADRI SUPER THERMAL POWER PROJECT

**LOCATION MANAGEMENT
INSTRUCTION**

LMI/ OGN/OPS/ELEC/039

Rev.0, August 2023

**Guidelines on Busduct
Maintenance and Overhauling**

APPROVED FOR

IMPLEMENTATION BY - - - - -

HEAD OF THE PROJECT (SIMHADRI)

Date: _____

NTPC LIMITED

LMI APPROVAL DOCUMENT

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3		Original/Ref. document	COS-ISO-00- OGN/OPS/ELEC/039 August 21
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Guidelines on Busduct Maintenance and Overhauling

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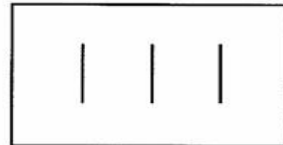
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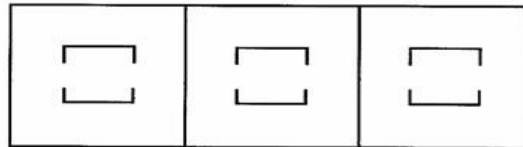
1.0 INTRODUCTION

There are several types of conductor arrangements used to transfer power from a source to its load. A metal enclosed bus is generally used to transfer power from generators to transformers/ transformers to switchgear. As illustrated in Figure 1, there are three basic types of metal enclosed buses:

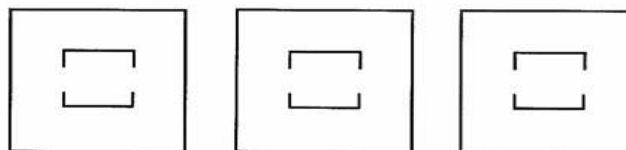
- Non-segregated phase busduct
- Segregated phase busduct (SPBD)
- Isolated phase busduct (IPBD)



Non-Segregated Phase Bus
with Insulated Flat Bar Conductors



Segregated Phase Bus
with Bare Channel Phase Conductors



Isolated Phase Bus
with Bare Channel Conductors

Figure 1 - Types of Busducts

1.1 Evolution of Busduct

In generator, short-circuit current in phase-to-ground faults could be limited by introducing a neutral impedance. However, phase to phase fault could not be limited by design. Thus, the bus design mainly focused on designing layouts that would prevent phase-to-ground faults from developing into phase-to-phase faults.

First design of *non-segregated phase bus*, was designed to limit the phase to earth faults due to rodents through a common metal cage/shield with conductors on insulators. This arrangement, works well if there are limited fault current magnitudes.

At larger loads, where voltages and short circuit magnitudes are high and load currents are over 5000 amperes, it is necessary to use tubular or channel conductors, both to carry the high load currents and to withstand the magnetic forces encountered during a fault. At larger loads, it has been customary to use only air insulation and to support the conductors on porcelain insulators. To prevent phase-to-phase faults with this construction, the phase conductors are spaced far enough apart so that metal barriers could be located between phases. Thus, the *segregated bus* resulted. One of the weaker points of the segregated phase construction was the possibility of the phase-to-ground arc burning through one of the metal inter-phase barriers, thereby developing into a phase-to-phase fault or double phase-to-ground fault.

Protection against burn-through is provided when each phase conductor is enclosed in its own metal housing and the three enclosures are then separated by air spaces. With this arrangement, even though an arc burns through the side of one enclosure, ionized gases cannot burn into adjacent enclosures. With this *isolated phase* construction resulted.

In isolated phase busducts, the conductor enclosures are electrically continuous and shorted together at both ends, circulating currents almost equal to the phase currents are induced in the enclosures, in opposite directions. An illustration of the circulating currents in continuous enclosures is provided in Figure 2.

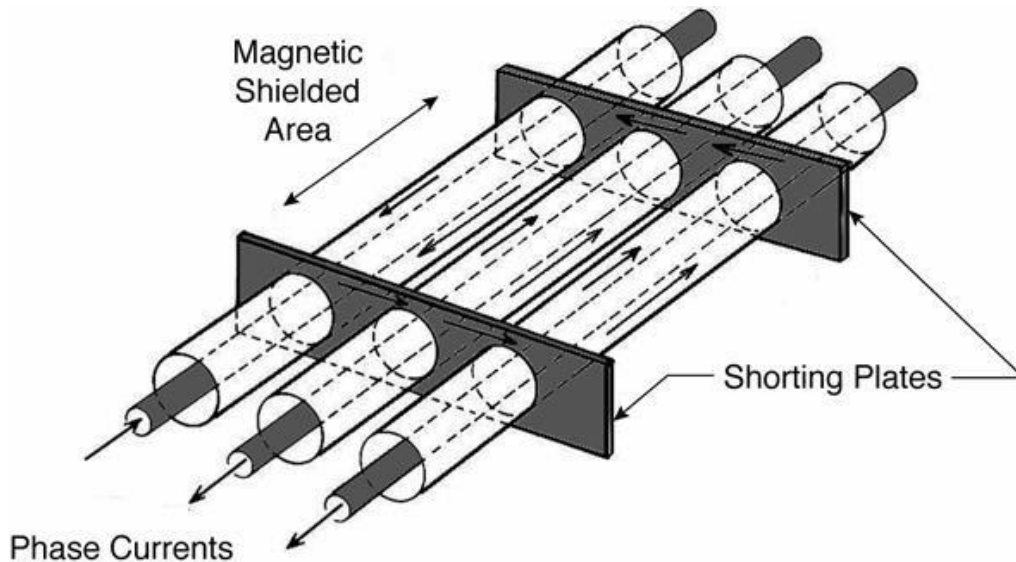


Figure 2 - Circulating Currents in Continuous Enclosures

1.2 Type of Busduct in NTPC stations

At NTPC stations, the connection between generator-to-generator transformers are through isolated phase busducts. The isolated phase busducts are continuous enclosure and self-cooled type. MV/HT busducts are mostly segregated type, however at some station non-segregated are also provided.

1. NTPC Simhadri stage-1, Generator busduct of IP Bus duct, make: BHEL-Jhansi, Rated voltage is 24KV of rated current 19KA/11KA/1.6KA
2. NTPC Simhadri stage-2, Generator busduct of IP Bus duct, make: C&S Electric Ltd, Rated voltage is 24KV of rated current 19KA/11KA/1.6KA

1.3 Bus Cooling and Busduct pressurization system

A busduct can be self-cooled, forced air-cooled, or water-cooled. In NTPC units, all busducts are self-cooled type.

1. Here in Simhadri: ELMECH make Bus duct air pressurization system for stage-1 and C&S Make Automatic air-drying bus duct pressurization system is employed for stage-2

A self-cooled bus relies on natural air circulation around the enclosures for cooling. Some air flow is desirable to reduce moisture accumulation during outage duration. When the entire bus is self-cooled, it is desirable to keep the system sealed, thereby keeping dirt and moisture out. Some self-cooled enclosures are kept slightly pressurized with dry instrument air to help prevent in-leakage of contaminants through busduct pressurization systems. Care must be taken to keep the pressure feed lines electrically insulated at each bus enclosure.

1.4 Seal-Off Bushings

Porcelain/Epoxy seal-off assemblies are supplied to prevent air flow from inside to outside bus sections. It is also generally provided in all ends of generator busduct wherever busduct pressurization system is envisaged. In some of the seal off bushings, current transformers (CTs) are also mounted.

In Simhadri Baroda bushings of 24KV epoxy resin cast type of Seal of bushings are installed.

1.5 Space heaters provided in Busducts and LAVT cubicles

The busducts and LAVT cubicles which are exposed to atmospheric air has been provided with space heaters & thermostat. The improper functioning of thermostat or space heaters may lead to faults due to moisture or burning due to localized heating.

2.0 SUPERSEDED DOCUMENTS - Nil**3.0 SAFETY INSTRUCTIONS**

All safety and general instructions laid down in the documentation supplied with the equipment must be followed at times by the maintenance personnel. Some of the general safety guidelines applicable to personnel and equipment are as follows:

(a) Personnel:

- Appropriate clothing, gloves, helmet, safety boots, harness etc.
- The personnel concerned must be familiar with the basic working regulations governing a given workstation: mechanical, dielectric, pressure hazards etc.

(b) Handling Equipment:

- This must be in good working order, regularly maintained, properly adjusted and compliant with international/national standards.

(c) Tools in General:

- Only use tools appropriate to the type of work to be carried out.

(d) Working Area:

- Make sure the floor is safe (free from oil, blunt objects, etc.).
- Make sure the confined spaces are properly ventilated before personnel enter.
- Obey the lock out /tag--out and permit to work procedures. Temporary grounds wherever used shall be monitored through permit or a monitored register.
- The site must be properly demarcated and kept clear
- Prior to starting work on generator busduct, it shall be isolated from Grid by opening isolators and disconnecting potential transformers & 100% SEF injection unit. It is to be ensured that AVR is properly isolated.
- Proper grounding of busduct & tools is to be ensured after isolation.
- Workers need to diligently monitor for changes in safe work clearances and employ test before touch when working on the electrical portions of the system
- While in service, the isophase bus carries 20-30 kA output from a nominal 15-24 kV generator. Electrical theory indicates that a conductor within the range of the magnetic field has the potential to become a lethal source of energy. Hence working in the vicinity of the isophase busduct in charged condition shall be done under area-engineer supervision only.

4.0 Maintenance Strategy of BUSDUCT

The maintenance strategy for the busducts must combine trending of key operating parameters, routine internal and external inspections, and consideration for age-related/life cycle management of subcomponents. This may require subcomponent refurbishment or replacement to meet the desired operating life of the plant.

4.1 Visual Inspection, Predictive Maintenance, and Performance Monitoring

The routine, periodic collection of visual observations, predictive maintenance data, as well as the collection of thermography data are the key inputs into a robust system monitoring plan.

4.1.1 Visual Inspections

Visual inspection of the bus duct should include support structures, expansion gasket/bellows, flexible bus jumpers and split covers (as applicable), bushing compartment, and ground connections.

Conditions that should be noted are peeling/blistering paint, crumbling/cracking of gaskets (from age or UV light exposure), corrosion/rust or other signs of water intrusion, overheating of gasket for corrosion (external signs of water intrusion), damage to structural supports, and bird nesting.

4.1.2 Thermography

Busducts are usually constructed with few organic materials. The material creep and the differences in expansion rates therefore dictate the maximum continuous operating temperature. Each manufacturer has developed specific limits depending on the design and materials used. The design limits are generally provided in busduct datasheet.

In-service, infrared (IR) inspection of an entire bus is a well-proven method of detecting enclosure hot spots and the heating of adjacent supports from stray circulating currents. Enclosure color and thermal emissivity are important. Outside the plant, direct sunlight can easily add 27°F (15°C) to enclosure temperatures. All aluminum surfaces should be painted to reduce corrosion and provide the thermal emissivity desired. Outside thermography inspections should not be done if there is direct sunlight on the bus. Solar radiation will result in false indications. Data collection is best performed at dawn, at dusk, or during an overcast day to obtain accurate data.

IR scanning should be performed at least once in each quarter, once with a high ambient temperature and once with a low ambient temperature. In addition, following an outage after the unit has reached full electric output, a survey should be performed.

During thermography data collection, visual observations and related performance parameters shall also be recorded. Items such as plant output, equipment line-up, local ambient temperature, busduct pressure may be required to properly diagnose anomalies.

These surveys are best performed when bus current is at a maximum. Overheating is increased as a square of the current. Usually, maximum megawatt and megavar loading coincide with both ambient temperature extremes. Severity criteria should be established for alert and alarm value. These values should be based on temperature rise of the material involved.

4.1.3 Partial Discharge

Air-insulated buses are a notorious source of partial discharge activity. Contaminated or cracked insulators, loose hardware, and slip-fit bus supports can generate high levels of electromagnetic interference (EMI) over a wide range of frequencies above 50 MHz.

Most partial discharge activity in the busduct does not always result in additional deterioration. Active sources can be trended. No bus is free of EMI, but a strong source indicates that a problem has developed that should be investigated. Comparing partial discharge investigation before and after overhauling may evaluate effective maintenance.

4.1.4 Off-Line Periodic Inspections

Internal components must be inspected when a bus is out of service. The entire bus, inside and out, should be visually inspected completely within 7-10 years. Even if no bus problems have been noted in the past, there is no assurance that trouble-free operation can be expected in the future.

Internal inspections are critical but must be scheduled to coincide with other important activities during those brief periods when a bus is out of service. Performing an inspection on different sections can be scheduled during every outage so that the entire bus is inspected within the recommended timeframe.

4.1.4.1 Foreign Material Exclusion (FME)

After safety, the next important consideration of maintenance of the busduct is foreign material exclusion (FME). FME has been cited as a cause of numerous operating experiences.

Therefore, finding FME and preventing FME issues from being generated by poor maintenance controls are keys to success. Control of tools and material that is taken in or brought out of the bus should be logged. Unexpected FME items found in the bus during inspection shall be removed and evaluated in the corrective action program. The time that inspection covers are open should be minimized to prevent exposure to dirt and moisture. The generator bushing box and transformer low-voltage bushing box should also be covered under the FME controls.

4.2 BUSDUCT Testing

Evaluation of a busduct system to determine suitability for service is performed both online and while out of service. Partial discharge tests can be applied only during operation. Most other tests must be scheduled for off-line application. A few, such as an overpotential test (HIPOT), are infrequently used. Others, such as an insulation resistance test, are often applied before a bus is returned to service at the end of each outage.

4.2.1 Insulation Resistance

An insulation resistance or Megger test should be a routine evaluation applied before a bus is returned to service. An isolated bus from associated equipment's with 5000-Vdc applied voltage will develop readings of over 1000 megohms when insulators are dry and clean. When the insulators are dirty and moisture is present, readings can be much lower.

A typical criterion for deciding if a bus is safe to energize is an insulation resistance between bus and ground of "1 megohm per rated kV plus 1 megohm" or greater. For instance, a 22-kV installation will have an acceptance criteria of 23 megohms or greater during a 5000-V Megger test. Insulation values below this value should be

carefully investigated. If a generator is connected to the bus, the machine insulation resistance will determine readings. With water cooled stators, resistance values will be very low due to the numerous paths to ground through stator bar water hoses. Values below 50,000 ohms have been measured on 24-kV systems when all three phases were parallel and bus, transformers, and generators were connected.

The use of dehumidifiers in busduct during scheduled outages will minimize the probability of unsatisfactory Megger and/or PI readings prior to unit restart.

4.2.2 Continuity

Low-resistance connections are required with a high-current bus. Measuring these micro-ohm joints is possible but difficult. Trending joint resistance is a good method to track deterioration.

Many areas are silver-plated, and contact resistance is, therefore, not directly related to bolting pressure. If an increase in joint resistance is measured, corrective action should be taken during that outage and not be postponed. Continuity of the inner conductor, outer tube, and all expansion joints can be compromised by cracks in the material or at welds. A detailed visual inspection should uncover these defects.

4.2.3 Overpotential/HIPOT

ANSI C37.20 provides information on overpotential testing of metal enclosed buses. Due to the use of porcelain insulators, ac test voltages are recommended over dc tests as information provided has little practical significance in determining suitability for service. Most insulators have long creep distance designs and will withstand rated impulse voltage wet or dry.

Maintenance overpotential testing is not generally performed. If a bus fault had occurred or major repairs were completed, an ac overpotential test would be needed. These test values could be either the shop test value or field test value, depending on the extent of repairs and confidence level desired.

OEM manual/datasheet may be referred for the testing values. If AC HIPOT performed during regular maintenance, then upto rated ac voltage should only be injected for 1 minute.

4.2.4 Tan-Delta Testing

Tan-Delta test shall be done at 10 kVac or rated bus voltage whichever is lower, to measure watts loss and power factor. This test is useful in trending, but ambient humidity and temperature may affect results.

However, a cracked insulator or general insulator contamination will usually show a significant change in trend results. A significant change from past readings may indicate that cleaning or further visual inspection is required.

4.3 Maintenance Task and Frequencies

The routine and regular maintenance activities shown in Tables 2-1 through 2-2 identify an effective maintenance program. Station to prepare LMI for site specific maintenance task list according to specific equipment installations, site operating experience and maintenance history.

Guidelines on Busduct Maintenance and Overhauling**Table 4-1 - Maintenance Task for Bus Duct**

Equipment	Work Description	Inspection Frequencies	Remarks
Bus duct (general)	General appearance of the bus duct and its associated systems (preservation, cleaning, and debris). Painting to be done every 10 years or earlier based on condition	Every Overhaul	Ensure aluminium end caps/blanking plates on generator and transformers ends during overhauling/major maintenance to avoid any foreign material entry. Minimum spares of busduct & its auxiliaries to be maintained.
	Proper clearance from all adjacent structures and equipment (permanent and temporary).		
	Proper connection between the bus duct enclosure and ground contact assembly and ground. Micro-ohm testing of all ground connections (disassemble, clean, and retest as required).		
	Proper bracing, suspension alignment, and evidence of damage to any of the bus duct structural supports. Inspect busduct insulators for cleanliness and any damage.	Every Overhaul	
	Exterior general cleanliness and/or nesting. Suitable rain hood arrangement shall be available on busduct outside the A row on all flange joints to avoid moisture ingress		
	Silica gel healthiness in breathers provided in non pressurized busduct (generally provided on both side of seal off bushing in SPBD)	Every month	
	Thermography of busduct	During SCC and after every 3 months as per CI 4.1.2	
	Partial discharge / EMI (electromagnetic interference) test to be planned such that rectification can be done during overhaul.	Before Overhauling	
	Proper mounting of the bus enclosure inspection covers and tightness.	During Overhaul	

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Equipment	Work Description	Inspection Frequencies	Remarks
	Tan delta test to be conducted at 10kVac or rated voltage whichever is lower		
	Investigation and corrective action if evidence of moisture/water coming from the inspection cover gasket area or standing water. Silicone sealant/putty to be applied for ensuring proper sealing.	During Overhaul	
	Interior cleanliness (free from moisture, dust, or foreign materials).	One-third of bus during overhaul.	
	Degradation of the inspection cover gasket. Replacement to be done whenever opened.	Complete busduct to be covered in every 10 years	
Insulators/ Seal off bushings	Check for evidence of porcelain chipping, cracking, arcing, flaking, missing hardware, cleanliness, or other physical damage.	One-third of bus during overhaul. Complete busduct to be covered in every 10 years	Perform insulation resistance tests. High-potential tests may be done when repair or replacement of parts/components has been performed. Refer manuals for high potential test values.
	Check for evidence of corrosion or discoloration due to heating.		
	Proper tightness of insulator mounting hardware. Gasket to be replaced during each inspection. Additional silicone sealant/putty to be applied for ensuring proper sealing.		
Bus conductors	Check condition of laminated links/bus connections, creep & cold flow of bolted terminations.	One-third of bus during overhaul. Complete busduct to be covered in every 10 years	Additional changes in termination as per Annex-I may be done if required.
	Healthiness of all washers and bolts to be ensured.		Belleville washers to be replaced during every overhaul
	Check clearance of flexible links outer surface from busduct housing more than minimum design clearance	During Overhaul	Drawing/data sheet may be referred for minimum clearance requirement.

Table 4-2 - Maintenance Task for Auxiliary Equipment's

Equipment	Work Description	Inspection Frequencies	Remarks
Current Transformer (CT)	Inspect CT and housing for signs of overheating, such as cracks, blisters, or discoloration.	Every Overhaul	
	Check healthiness of conduit connecting the CT secondary lead boxes		The conduit should be either non-metallic or non-magnetic.
	Ensure the healthiness of wires from CT terminal to intermediate JB		Spring washers should be used in

Guidelines on Busduct Maintenance and Overhauling

Equipment	Work Description	Inspection Frequencies	Remarks
	and to CTJB are properly routed. Conduct insulation resistance test.		busduct mounted CTJB.
	Conduct CT testing like resistance, magnetization test and megger. Measure CT resistances and compare with earlier/commissioning values.		Make checklist (OIN-OPS-ELEC-032 may be referred)
	Thermography of CT JB's	During SCC and every quarter at around full load	
Rubber expansion boots and bellows	Inspect for material condition, dryness, cracks, or signs of air leakage.	Every overhaul	Replacement to be planned every 10 years
Air pressurization system	Check functioning of dryers and complete system/changeovers etc. Busduct Pressurization system alarm to be provided in control room.	Monthly	Make checklist as per system
	Check the dew point of air at inlet & outlet of system	Monthly	Records the values
	Check if the air pressure being maintained in busduct.	Daily	To be included in daily log sheet of operation
	Check air tightness test of busduct. Additional sealing to be done through silicone sealant/putty etc	During overhaul	Allowable air leakage shall be less than 5% of total enclosure volume per hour
Busduct Space heaters	Space heaters current to be measured	Every month	Record to be maintained
	Thermography check. Functioning of heater may be checked during thermography when the ambient temperature is low.	Every three months	
	Space heaters and thermostat to be checked for proper functioning	Every overhaul	Thermostat may be set for cut-in at ard 30 deg and cut out at ard 55 deg
Potential Transformer (PT)	Check PT connection to bus for signs of corrosion/ Overheating/creep.	Every overhaul	All PTs to be replaced every 20 years or based on condition. This supersedes all earlier circulars
	Inspect fuse clip and secondary contact alignment and silvered connections. If limit switch is kept in service, then check for healthiness.		Limit switch to be replaced every 5 years
	Inspect sliding ground contact and bus. Gen busduct PT primary earth		

Guidelines on Busduct Maintenance and Overhauling

Equipment	Work Description	Inspection Frequencies	Remarks
	should be provided directly from earth through sufficient loop for rack in-out of PT.		
	Proper alignment to be checked while rack-in of Generator PTs. Proper locking of PT panel door of switchgear/generator PT cubicles.		Lock to be changed if found damaged
	Inspect all mechanical and electrical connections. Inspect all insulators for any sign of tracking/moisture		
	Inspect/replace PT primary and secondary fuses. Check functioning of MCB's and tightness of secondary connections.		Replacement of fuses after every 5 years and MCB after 10 years
	Check the functioning of PT cubicle space heater and thermostats.		Thermostat may be set for cut-in at ard 30 deg and cut out at ard 55 deg
	Conduct PT tests (Ratio, IR, magnetizing). Check the secondary burden resistance "PT continuity check".		Make checklist (OIN-OPS-ELEC-032 may be referred)
	Take voltage readings at the secondary side during unit running.	During SCC/OCC at local JB and Relay panel.	Ensure records
Surge arrestor	Record leakage current and counter readings daily through Operation log sheet	Daily	Change in readings to be investigated by maintenance.
	Check Insulation resistance. Physical inspection including pressure relief plates	Every overhaul	All surge arrestor to be replaced every 20 years or based on condition. This supersedes all earlier circulars
Surge capacitor	Check Tan delta and capacitance at 10kVac, IR values	Every overhaul	All surge capacitor to be replaced every 20 years or based on condition. This supersedes all earlier circulars
	Physical inspection for oil leakage and insulator damage.		

5.0 EXCEPTION REPORTING & DISPENSATION OF DEVIATION

Any deviation would be reported to Head of EMD

6.0 Responsibility

Area engineer of concern area is responsible for implementation and overall responsibility is lies with the sectional heads and record keeping as per Annexure-III task list.

7.0 Review

This LMI document will be reviewed once in 2-years by head of the project NTPC Simhadri.

Annexure-I

DETERIORATION AND RESTORATION

A) Various operating issues due to Bus Terminations and Gaskets

There are links between the busduct and the generator phase lead bushings, busduct and transformer bushings, neutral bus connections, and between sections of the busduct itself. The insulation gaskets are also provided in between busduct sections.

Issues include the following:

- A broken/damaged braided flexible links.
- Cracked bolts due to repetitive over-torque to bolt values.
- Creep/cold flow of bolt holes due to repeated torquing.
- Over-torquing or under-torquing of electrical connections.
- Deterioration of silver or tin plating on bolted joints.
- Loose connections due to improper hardware/mismatch of flat and Belleville washers.
- Loose/incorrect connection due to change in orientation of the flexible link.
- Incorrect installation of insulating parts or use of undersized insulating parts that results in burning/sparking damage inside the generator lead/neutral box.
- Localized heating due to loose/failed ground straps can cause loss of gasket seal and loss of pressure boundary.
- Crumbling, cracking, and hardening of gaskets on inspection doors, permitting water to enter the bus
- Deterioration of rubber expansion joints (boots) from heat and ultraviolet (UV) exposure
- Peeling and flaking of paint, exposing bare aluminum and resulting in higher temperatures
- Drain holes filling with dirt, permitting water accumulation

When dealing with parallel current paths, it is useful to remember that overheated sections are carrying higher load than the cooler sections (with higher resistance connections). The connections that appear to be cooler are the ones that require tightening.

B) Bolting Considerations

High-resistance connections in a high-current-carrying system create thermally damaging overheating conditions.

Use of stainless-steel bolting is recommended due to minimum effects from thermal cycling. Belleville washers are also recommended for this reason. Use of a stainless-steel flat bar backing plate with tack-welded nuts can be used for ease of installation. Additionally, the more uniform compression of the current-carrying surfaces will maximize the area available to conduct current. It is important to note that Belleville washers should not be reused.

C) Flexible Links

Flexible braid-style links can also be damaged by improper maintenance techniques. Electricians need to be careful not to overstress the braids by bending them backward during removal and installation. Repeated flexing or bending could result in breaking of strands and greatly reduced life. OGN-OPS-ELEC-035 shall be referred.

D) Split Covers

The ends of continuous enclosure bus are fitted with thick shorting plates welded to all three of the outer tubes. Beyond this shorting plate, a very strong magnetic field exists. This is often the case at transformers; however, near the generator bushings, split covers are sometimes used (as shown in Figure A-1) to eliminate the external magnetic field and reduce mechanical forces on the generator bushings during a bus fault. This assembly has all of the features of the main bus except that it is insulated from adjacent structures and grounded at only one location.

Secondary grounds can result in unwanted circulating currents and overheating of adjacent structures or the split covers.

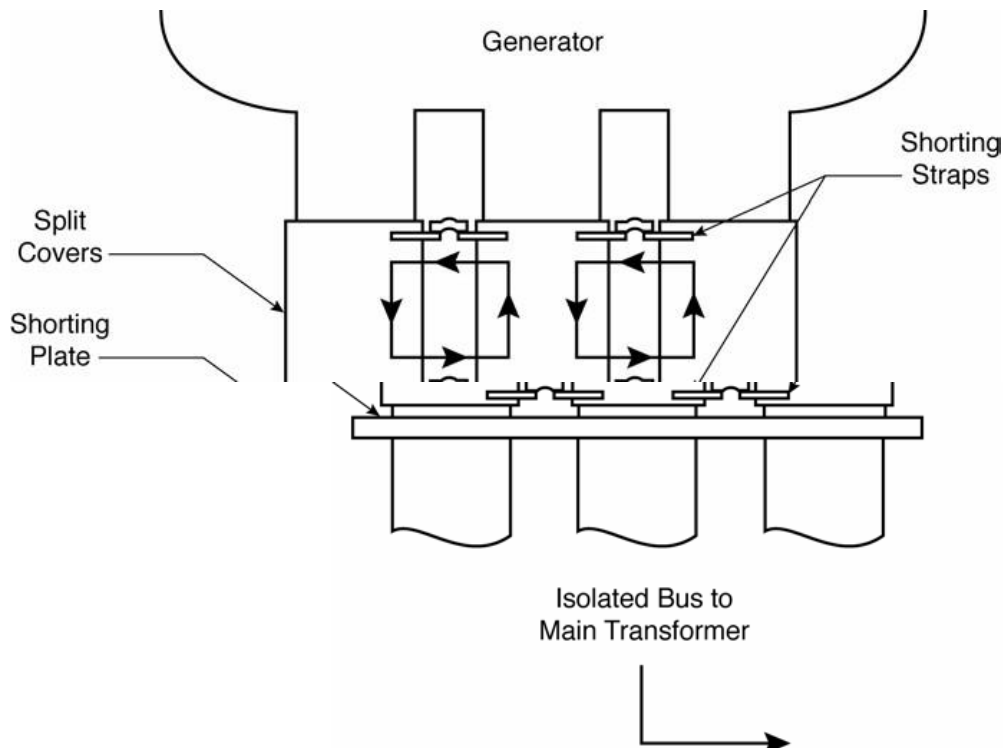


Figure A-1 Typical Arrangement of Split Covers

Heavy shorting straps are fitted at both ends of the split covers to carry the circulation current. Mechanical misadjustment is a frequent problem. All three split covers must be positioned correctly to properly attach the shorting straps. Distortion of components results when final alignment is forced.

E) Insulation Gaskets

Failure of thin insulation gaskets at busduct ends may also lead to bus faults. Replacement of all original gasket material with insulation rated for Class H, 356°F (180°C) operation has proven to be a long-term solution to deterioration.

F) Inspection Covers

Removable hatches or inspection covers are often provided along the enclosure to permit access to center conductor support insulators, expansion joints, and other internal components. These covers should be both air and water tight to keep contamination out of the bus. Gasket material often hardens and cracks because of fluctuation in operating temperatures. Same is to be replaced or sealed additionally with silicone sealant.

G) Rubber Expansion Boots

With many designs, each enclosure is fitted with a rubber expansion joint. These “boots” keep rain and dirt out. An insulating material is necessary to eliminate stray currents in this non-shielded area. Usually, the original material deteriorates from heat and sunlight in 15–20 years. Replacements to be planned after every 15 years of service. If non-OEM boots are installed, no wire-reinforced material can be used and only high-temperature (Thermal Class F, 311°F [155°C]) components will provide long life.

Annexure-II

GOOD PRACTICES FOR ELECTRICAL CONNECTIONS

The same problems and approach apply to all electrical connections throughout the plant.

1) Basic Philosophy and Facts

- All voltage and current are meant to pass between the contact surfaces of the conductors and not through the bolting.
- Maximum contact surface between the conductors is essential to a good electrical connection.
- Conductors are generally copper, brass, or aluminum, while bolting is generally steel.
- Steel bolting is stretched slightly by tightening the nut on the bolt.
- Copper, brass, and aluminum will “creep,” “conform,” “expand,” or “move away” from stress imposed by steel bolting being stretched.
- The flexible links between bus segments are removed and replaced periodically to facilitate electrical testing of the generator.
- The solid-appearing ends of flexible links are not solid; rather, they are the flexible link squeezed into a rectangular shape with a copper wrap around the outside.
- Dimples or concave shapes occur in the ends of flexible links or copper or aluminum conductors because of “creep” of the relatively soft material as compared to the steel bolting material.
- A flexible link that is darkened due to high current is the best conductor. The other flexible links have poor electrical connections and thus high resistance at the connection point, which causes the current to pass through the flexible link with the good electrical connection.
- An electrical circuit can “ring” or light a bulb to ensure continuity, but this is not an indication of the current-carrying capabilities of the connection. A poor electrical connection heats up, which causes a worse connection that produces more heat. Micro-ohm testing is recommended to ensure that quality connections have been made.
- Cleaning of plated areas should be performed with denatured alcohol and clean, dry, lint-free cloths or pressurized air. The use of Scotch-Brite® pads should be discouraged.

2) Solutions to “Creep” of Electrical Components

- Fabricate and install stainless steel plates on the backside of conductors to spread out the compression imposed by the bolting.
- Use Belleville washers on bolting to spread load and provide compression that will accommodate “creep” of conductors. Never reuse Belleville washers.
- Check all large electrical connections with a 0.002-in. (0.05-mm) feeler gauge to ensure surface contact.

Annexure-III
COS-ISO-00-OGN/OPS/ELEC/039, Rev. No. : 0.0
Guidelines on Busduct Maintenance and Overhauling

Maintenance Task list

Equipment	Work Description	Inspection Frequencies	Remarks	Done/Not done
Bus duct (general)	General appearance of the bus duct and its associated systems (preservation, cleaning, and debris). Painting to be done every 10 years or earlier based on condition	Every Overhaul	Ensure aluminum end caps/blanking plates on generator and transformers ends during overhauling/major maintenance to avoid any foreign material entry. Minimum spares of busduct & its auxiliaries to be maintained.	
	Proper clearance from all adjacent structures and equipment (permanent and temporary).			
	Proper connection between the bus duct enclosure and ground contact assembly and ground. Micro-ohm testing of all ground connections (disassemble, clean, and retest as required).			
	Proper bracing, suspension alignment, and evidence of damage to any of the bus duct structural supports. Inspect busduct insulators for cleanliness and any damage.	Every Overhaul		
	Exterior general cleanliness and/or nesting. Suitable rain hood arrangement shall be available on busduct outside the A row on all flange joints to avoid moisture ingress			
	Silica gel healthiness in breathers provided in non pressurized busduct (generally provided on both side of seal off bushing in SPBD)	Every month		
	Thermography of busduct	During SCC and after every 3 months as per CI 4.1.2		
	Partial discharge / EMI (electromagnetic interference) test to be planned such that rectification can be done during overhaul.	Before Overhauling		
	Proper mounting of the bus enclosure inspection covers and tightness.	During Overhaul		

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Equipment	Work Description	Inspection Frequencies	Remarks	Done/Not done
	Tan delta test to be conducted at 10kVac or rated voltage whichever is lower			
	Investigation and corrective action if evidence of moisture/water coming from the inspection cover gasket area or standing water. Silicone sealant/putty to be applied for ensuring proper sealing.	During Overhaul		
	Interior cleanliness (free from moisture, dust, or foreign materials).	One-third of bus during overhaul.		
	Degradation of the inspection cover gasket. Replacement to be done whenever opened.	Complete busduct to be covered in every 10 years		
Insulators/ Seal off bushings	Check for evidence of porcelain chipping, cracking, arcing, flaking, missing hardware, cleanliness, or other physical damage.	One-third of bus during overhaul. Complete busduct to be covered in every 10 years	Perform insulation resistance tests. High-potential tests may be done when repair or replacement of parts/components has been performed. Refer manuals for high potential test values.	
	Check for evidence of corrosion or discoloration due to heating.			
	Proper tightness of insulator mounting hardware. Gasket to be replaced during each inspection. Additional silicone sealant/putty to be applied for ensuring proper sealing.			
Bus conductors	Check condition of laminated links/bus connections, creep & cold flow of bolted terminations.	One-third of bus during overhaul. Complete busduct to be covered in every 10 years	Additional changes in termination as per Annex-I may be done if required.	
	Healthiness of all washers and bolts to be ensured.		Belleville washers to be replaced during every overhaul	
	Check clearance of flexible links outer surface from busduct housing more than minimum design clearance	During Overhaul	Drawing/data sheet may be referred for minimum clearance requirement.	

Maintenance Task for Auxiliary Equipment's

Equipment	Work Description	Inspection Frequencies	Remarks	Done/Not done
Current Transformer (CT)	Inspect CT and housing for signs of overheating, such as cracks, blisters, or discoloration.	Every Overhaul		
	Check healthiness of conduit connecting the CT secondary lead boxes		The conduit should be either non-metallic or non-magnetic.	
	Ensure the healthiness of wires from CT terminal to intermediate JB		Spring washers should be used in	

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Equipment	Work Description	Inspection Frequencies	Remarks	Done/Not done
	and to CTJB are properly routed. Conduct insulation resistance test.		busduct mounted CTJB.	
	Conduct CT testing like resistance, magnetization test and megger. Measure CT resistances and compare with earlier/commissioning values.		Make checklist (OIN-OPS-ELEC-032 may be referred)	
	Thermography of CT JBs	During SCC and every quarter at around full load		
Rubber expansion boots and bellows	Inspect for material condition, dryness, cracks, or signs of air leakage.	Every overhaul	Replacement to be planned every 10 years	
Air pressurization system	Check functioning of dryers and complete system/changeovers etc. Busduct Pressurization system alarm to be provided in control room.	Monthly	Make checklist as per system	
	Check the dew point of air at inlet & outlet of system	Monthly	Records the values	
	Check if the air pressure being maintained in busduct.	Daily	To be included in daily log sheet of operation	
	Check air tightness test of busduct. Additional sealing to be done through silicone sealant/putty etc.	During overhaul	Allowable air leakage shall be less than 5% of total enclosure volume per hour	
Busduct Space heaters	Space heaters current to be measured	Every month	Record to be maintained	
	Thermography check. Functioning of heater may be checked during thermography when the ambient temperature is low.	Every three months		
	Space heaters and thermostat to be checked for proper functioning	Every overhaul	Thermostat may be set for cut-in at ard 30 deg and cut out at ard 55 deg	
Potential Transformer (PT)	Check PT connection to bus for signs of corrosion/ Overheating/creep.	Every overhaul	All PTs to be replaced every 20 years or based on condition. This supersedes all earlier circulars	
	Inspect fuse clip and secondary contact alignment and silvered connections. If limit switch is kept in service, then check for healthiness.		Limit switch to be replaced every 5 years	
	Inspect sliding ground contact and bus. Gen busduct PT primary earth			

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Guidelines on Busduct Maintenance and Overhauling

Equipment	Work Description	Inspection Frequencies	Remarks	Done/Not done
	should be provided directly from earth through sufficient loop for rack in-out of PT.			
	Proper alignment to be checked while rack-in of Generator PT's. Proper locking of PT panel door of switchgear/generator PT cubicles.		Lock to be changed if found damaged	
	Inspect all mechanical and electrical connections. Inspect all insulators for any sign of tracking/moisture			
	Inspect/replace PT primary and secondary fuses. Check functioning of MCB's and tightness of secondary connections.		Replacement of fuses after every 5 years and MCB after 10 years	
	Check the functioning of PT cubicle space heater and thermostats.		Thermostat may be set for cut-in at arid 30 deg and cut-out at arid 55 deg	
	Conduct PT tests (Ratio, IR, magnetizing). Check the secondary burden resistance "PT continuity check".		Make checklist (OIN-OPS-ELEC-032 may be referred)	
	Take voltage readings at the secondary side during unit running.	During SCC/OCC at local JB and Relay panel.	Ensure records	
Surge arrestor	Record leakage current and counter readings daily through Operation log sheet	Daily	Change in readings to be investigated by maintenance.	
	Check Insulation resistance. Physical inspection including pressure relief plates	Every overhaul	All surge arrestor to be replaced every 20 years or based on condition. This supersedes all earlier circulars	
Surge capacitor	Check Tan delta and capacitance at 10kVac, IR values	Every overhaul	All surge capacitor to be replaced every 20 years or based on condition. This supersedes all earlier circulars	
	Physical inspection for oil leakage and insulator damage.			